

What It Now Claims, and What Would End It

*The whole framework in its current position: six symbols at one end, a gauge group and a spectrum at the other, and four ways it can be killed.
Written to be attacked.*

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BEFORE ANYTHING ELSE

The four ways this dies

A framework that leads with its results is asking to be believed. A framework that leads with its throat is asking to be tested. The second is the only request worth making, so the kill conditions come first, before a single claim is stated. Each is specific, each is near-term, and none has an escape written for it.

A single-deed pipeline returning w not equal to minus one The framework says the vacuum readout is a constant boundary record, so the dark energy equation of state is exactly minus one and does not evolve. Apparent evolution is permitted only as an artefact of summing pipelines of different closure count. If DESI's final release shows evolving dark energy in BAO alone, with no supernova or CMB pipeline attached, there is no artefact left to appeal to and the framework is finished. This is the nearest and sharpest of the four.

A fourth light chiral generation The family index splits one plus three under the decay's own Weyl group, and the criterion that selects three is stated. A fourth light generation falsifies it. The one plus three splitting is character theory and cannot be renegotiated; only the count is at stake, and the count is three.

A confirmed low-energy gauge structure outside $so(10)$ The surviving algebra is forced once the decayed subsystem is fixed. $so(10)$ contains $su(5)$ and therefore the Standard Model, so this is not an easy kill, but anything genuinely outside it ends the route.

A running self-holding system that fails to return quantised retention at powers of r Under a convention declared before the count. This has been the framework's own stated condition of defeat for three printings and it has never been run. It cannot be settled at a desk and it is the one item on this list that nobody has yet attempted.

Two secondary stakes are live and should be named with them. Dark matter must carry no visible gauge charge. And the observer boundary must have genus zero, or Gauss-Bonnet returns four pi times one minus the genus and the vacuum readout changes or vanishes. Both were published before the numbers that now depend on them.

SECTION ONE

The spine, in one page

The framework begins with a sentence asked to describe itself and ends with a gauge group. The chain is short enough to print, and its shortness is the point: there is no long sequence in which an error could be hiding.

phi squared equals phi plus one

Six symbols. From the two-sided reading of a self-description, the retained fraction is one over two phi. From the update's minimal form, the stable ratio is the golden fixed point. From the two channels squared, three weights that exhaust the whole. From the discreteness of closure, a crossing price that is exactly r . From the same two-sided reading, a carrier that equals its own dual, even and positive definite, which at rank eight is E8 and nothing else.

Then the descent. A reduction that preserves the system leaves a readable algebra, and that algebra is the physics. What survives is three modes of external distinction, one local clock and one closure, and the opposite roles of the first two force a Lorentzian signature. The rest of the corpus is the consequences: superposition, complex phase, the Born pricing, unitarity, gravity as the surviving part of carrier nonclosure, and mass as the cost of persistence.

The certified dependency graph over that whole structure has sixty-four nodes and sixty-seven edges, is acyclic, and its longest chain is seven steps. The newest result in the corpus, a candidate grand unified group, sits five steps from the six symbols above. The graph ships as source with every edge carrying the phrase that licenses it.

SECTION TWO

What was closed in the Fifth Printing and after

The corpus carried seven proof obligations for two printings. They did not move, and the reason they did not move is instructive: the Register said most of them waited on the first, so every attempt began there and bounced. Attacking from the other end determined the first rather than depending on it. The arrow in the ledger had been pointing the wrong way.

The rank-three decay

Given that the clock and closure modes are each single, both already proved, the claim that the spatial modes number three is identical to the claim that the surviving block has dimension five, and therefore that the decayed part has rank three. E8 has exactly three rank-three subsystems up to conjugacy, and the enumeration is exhaustive because the Weyl group is transitive on the roots. Two of the three are then excluded by results the corpus already proves: one hosts no closed triangular history, the other is reducible and would require two decay operators where the corpus derives its sectors from one.

The survivor is unique. The reduction is averaging over a group of order twenty-four, the gauge algebra is $so(10)$, matter sits in the spinor sixteen with a vector ten beside it, and the two hundred and forty-eight sorts as fifteen plus forty-five plus one hundred and twenty-eight plus sixty with nothing left over. The sixteen of $so(10)$ is one generation of Standard Model fermions with a right-handed neutrino included rather than appended.

Three spatial dimensions and $so(10)$ are the same fact. The corpus had been carrying them as two separate unknowns.

Three families

The family index is the four of the decayed algebra, and as a module for the group the reduction averages over it is the permutation module on four points, which splits as one plus three. The invariant direction is the one the decay cannot tell apart from itself; it connects no distinct sector and produces no separate readable identity. Three remain.

The complete carrier closure

A closure passing each of the two hundred and forty roots exactly once exists, by explicit construction, with a witness verified independently of the search that found it. The closing condition turned out to be free: roots come in opposite pairs and sum to nothing. The standing vacuum readout therefore rests on a proved lemma rather than a granted one.

Chirality

The published no-go theorem for theories of everything inside E_8 assumes an $SL(2, \mathbb{C})$ subgroup with the gauge group as its centraliser. This framework never builds that. Its decay group is finite of order twenty-four and fixes the surviving block pointwise, and its Lorentzian structure is an inherited form on surviving directions rather than a gauged subgroup. The theorem remains correct and remains fatal to the models it addresses; this construction is not among them.

Escaping the theorem produces nothing on its own, so the mechanism was built. Matter is a cycle, cycles have direction, and reversal exchanges the sixteen with its conjugate, which was checked across all one hundred and twenty-eight roots of the sector. The record accumulates one way and joins rather than overwrites, both proved. So of a cycle and its reverse only one reproduces a readable identity, and the conjugate branch is that same matter read against the arrow, which is what an antiparticle has meant for eighty years.

The protection is the sharper half. A mass term joining the sixteen to its conjugate composes a cycle with its own reverse, and the net displacement is exactly zero: a closed traversal that writes no record. That is the configuration the corpus has forbidden since the wound movements, under the name overwrite rather than retention. The term that would make an E_8 spectrum vector-like is the term the framework already banned, and it banned it to generate the arrow of time.

SECTION THREE

The variational principle was already written

The corpus has been saying for two printings that it owed a dynamical law. It did not. Movement Nineteen states that the square of a defect formed from curvature weighted by u against realised area weighted by r has three terms and no others. Write them out.

$$D = u R + r (e \wedge e)$$

$$D \wedge D = u^2 (R \wedge R) + 2ur (R \wedge e \wedge e) + r^2 (e \wedge e \wedge e \wedge e)$$

That is MacDowell-Mansouri gravity term for term: a topological Gauss-Bonnet piece, an Einstein-Hilbert piece, and a cosmological constant piece. And the identification is overdetermined, which is what

makes it a result rather than a resemblance. Two independent coefficient ratios must return the same cosmological constant, and there is nothing available to tune because u and r are pinned.

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from the cross term    Lambda = 1.341640786499874
from the quartic term  Lambda = 1.341640786499874
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$\Lambda = 3r/u = 3 \text{ over root five, exactly}$

Three arbitrary coefficients would not be consistent with a single cosmological constant. These are, because they are the expansion of a square, and the square is the framework's own bookkeeping identity. The largest of the three weights sits on the topological term, which in four dimensions does nothing locally: the biggest thing in the action is the part that cannot be felt.

PROVED as the identification and the closed form; certified by scripts/lagrangian.py. PROPOSED as the claim that this is the framework's complete dynamics rather than its gravitational sector alone. What the framework still lacks is a matter action, and Section Five records that.

SECTION FOUR

Hydrogen, from the carrier

The three modes are traversable and carry a closure, so every traversal in them returns, and the world is single so the space is simply connected. A closed simply-connected three-manifold is the three-sphere; that is Perelman's theorem and the framework borrows it with its name attached.

There was an objection to be met here and it dissolves on inspection. The three-sphere on which hydrogen's bound states live is a momentum-space sphere, and the framework's would be a position-space one, so the identification looks like an equivocation. It is not, because the carrier equals its own dual. That is what the two-sided reading demands and what E8 delivers, and on a self-dual carrier there is no position-against-momentum distinction to make. One sphere, read two ways, which is the framework's oldest move applied to itself.

Then Fock's construction runs. Bound states are harmonics of degree n minus one, the eigenvalue condition fixes the compactification scale as one over n, and two things follow with no atomic input whatever.

n	N=n-1	dim H_N	n^2	ell decomposition	
1	0	1	1	s	(1)
2	1	4	4	sp	(1 + 3)
3	2	9	9	spd	(1 + 3 + 5)
4	3	16	16	spdf	(1 + 3 + 5 + 7)
8	7	64	64	spdfghik	(1 + ... + 15)

The n squared degeneracy, the entire shell structure of s, p, d and f, and the inverse-square law in n, all from a harmonic count. The Balmer ratios follow without any coupling entering at all. The Rydberg itself needs the coupling and is not claimed.

One correction to the hydrogen monograph belongs here. Its computed cutoffs, exact through n equals three on the 24-cell and n equals six on the 600-cell, are the spherical design strengths of those polytopes, five and eleven respectively, with the maximum harmonic degree being the floor of half the strength. Both cutoffs, exactly. A predicted result outranks an observed one and the monograph should say so.

SECTION FIVE

The honest ledger

What the framework owes, with the reason each item stands, and with the items that moved marked as moved.

One lattice condition The reduction is forced to be an average over a reflection subgroup provided the decayed sublattice is generated by roots. Everything else in that argument is standard: a norm-one projection onto a subalgebra is a conditional expectation, and the covariant conditional expectation onto a finite-group fixed algebra is unique and is the average. But rootless rank-three sublattices of E8 are the common case, not the exception, so the condition is genuine. The corpus's own primitive-crossing definition argues for it, since a subspace holding no crossing hosts no history and so has nothing that could have decayed. Reduced from an open question about operator algebras to one finite condition. OPEN.

One chirality argument, halfway The orientation mechanism is built and the protection is identified. The bridge from a forbidden traversal composition to a term in a Lagrangian is PROPOSED and is the load-bearing step of Paper 22. It cannot be made rigorous without a matter action.

One matter action Section Three supplies the gravitational sector. There is no corresponding statement for the fermions or the gauge fields, and until there is, the chirality protection is structural rather than derived. This is now the largest single gap in the framework.

One coupling The fine structure constant, or equivalently the Rydberg's scale, or equivalently the normalisation of the Einstein term. The same debt appears in three costumes throughout the corpus and is owed once.

One anomaly calculation In modules that are now named, so it is arithmetic rather than an open form.

One closure count Whether the complete carrier closure is unique and how many exist. Not required by anything the corpus claims, since every complete closure has two hundred and forty crossings by definition. OPEN and finite.

Two measurements The pipeline audit of Section Six, and the running self-holding system. Neither closes at a desk.

One reading, one bridge, one action, one coupling, one calculation, one count and two measurements. That is the whole of it. Everything else from the first commitment to the newest result is proved, conditional on a condition written into its own statement, or dissolved as a unit convention.

SECTION SIX

The dark energy comparison, audited by the corpus's own rule

The rule is Paper 19's and it must be obeyed in order: count the whole-body closures each pipeline performs, declare the difference, and only then consult the gap. Counts are read off architecture.

CMB	light forms, is released, crosses the structure era, closes through a model engine converting spectra to parameters	3
BAO	a ruler fixed at the drag epoch, then read locally in matter tracers; formation inherited, nothing else	1
SNe Ia	parallax to Cepheid to supernova, local reads throughout	0

CMB vs SNe Ia	dn = 3	retained 0.914061	gap 8.594%
BAO vs CMB	dn = 2	retained 0.941854	gap 5.815%
BAO vs SNe Ia	dn = 1	retained 0.970492	gap 2.951%

The first line is Paper 19's Hubble prediction, reached from the other end by the same architecture, which is a consistency check the corpus had not performed. Now the published significances for evolving dark energy, by combination.

DESI DR2 BAO alone	no preference for evolution
DESI + CMB	3.1 sigma
DESI + CMB + Pantheon+	2.8 sigma
DESI + CMB + Union3	3.8 sigma
DESI + CMB + DESY5	4.2 sigma

The single-deed pipeline shows nothing. The three-deed fits differ from one another by more than one and a half standard deviations while using identical baryon acoustic and microwave data; only the supernova pipeline changed. A physically evolving equation of state cannot care which photon pipeline is attached to it. A priced difference between deeds must.

The Observational Standard already carried this as a standing audit note before the question was put, in the entry recording that different supernova samples affect the evolving-dark-energy significance. The framework was watching the right quantity.

Consistent with the framework's reading and explicitly NOT confirmation of it. The map from a priced deed difference onto the two parameters of an evolving equation of state is not derived here and is OPEN. What the reading buys is a prediction about the future of the anomaly: a physical evolution tightens toward one value as statistics improve, while a deed difference keeps tracking which pipelines were summed. Those diverge, and DESI's final release separates them.

SECTION SEVEN

What is being claimed, stated flatly

Proved, from stated commitments: the forced order of the seven doings, the uniqueness of the minimal update, the golden fixed point, the retained fraction, the three weights, the exactness of the crossing price, the triangular minimal history, the root cell area, the golden ring pairing, the kernel characterisation of the physical algebra, the distinctness of minimal and general decay generators, the singleness of the clock and closure modes, superposition, complex phase, the Born pricing, unitarity, the identity of the boundary weight with the interference term, retention over replacement, the singleness of records, the exhaustive enumeration of rank-three subsystems, and the existence of a complete carrier closure.

Conditional, on conditions written into their own statements: that the carrier is E8, that the inherited form is Lorentzian, that the decay is on A3 with surviving symmetry so(10), and with it the spatial rank, the matter modules and the family count.

Proposed, as physical readings offered for falsification: that the cross-term of the gravitational expansion is ordinary gravitational dynamics, that the standing vacuum readout is the cosmological constant, that the sorting of parity by block reversibility is physical, and that the forbidden traversal composition is the field-theoretic mass term.

Not claimed anywhere: that these readings are correct, that the carrier is the right carrier, that agreement with a measured value is a derivation of it, or that any sentence in the corpus is a completed theory of quantum gravity.

The dependency graph shows that no result in the corpus depends on any open obligation. The obligations are terminal. The argument is closed and what remains is a forward work programme, and that is a checkable property rather than a claim about mood.

SECTION EIGHT

What would make this a theory rather than a framework

Three things, in the order they should be attempted, and the ordering is not arbitrary: each unlocks the next.

The matter action Section Three supplies gravity. The same square that produced it has not been applied to the sectors result twenty-five classifies. If the three weights coefficient the fermion and gauge terms the way they coefficient the gravitational ones, the chirality bridge becomes a derivation and the coupling problem becomes one calculation rather than three.

The coupling With a matter action, the fine structure constant is a normalisation rather than an input, and the Rydberg follows, and hydrogen closes completely rather than in form only.

One number, before it is measured Everything in this corpus either reconstructs known structure or matches a known value. The vacuum readout at a fraction of a standard deviation is a postdiction and costs nothing to be right about. The framework will not be taken seriously on reconstruction, however elegant. The dark energy stake of Section Six is the closest thing available and it is a prediction about the behaviour of an anomaly rather than a number. A number is still owed.

And then the thing that no amount of this substitutes for. Every audit in this corpus has been internal: the framework checked against itself, by its own instruments, in its own frame. That is now done to a standard most published work does not meet, and it is not the same as contact. What the corpus can honestly say is that it is finally in a condition where contact is possible, because every number regenerates by one command, every claim carries a grade, every open item states why it is open, and the four kill conditions at the front of this paper are specific enough to act on.

it is ready to be wrong in public, which is the only kind of ready that counts